The binomial case of Graham's conjecture on polynomial representations with prescribed sum of reciprocals

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Abstract

Let $f(x)$ be an integer polynomial with positive leading coefficient and such that $\gcd(f(1), f(2), \ldots) = 1$. Graham conjectured that for every positive rational α an $n_0 = n_0(f(x), \alpha)$ exists such that for all $n \geq n_0$ distinct positive integers $a_1, \ldots, a_r$ exist with $f(a_1) + \cdots + f(a_r) = n$ and $\frac{1}{a_1} + \cdots + \frac{1}{a_r} = \alpha$. For a polynomial $f(x) = ax^d + b$ and any $\alpha > 0$, we prove that the existence of such an n_0 can in principle be reduced to a finite computation. For $\alpha = 1$ we thereby confirm Graham's conjecture for a few thousand linear and quadratic polynomials.

1 Introduction and notation

For a polynomial $f(x) : \mathbb{Z} \to \mathbb{Z}$ and a positive rational α, let $n_0 = n_0(f(x), \alpha)$ be the smallest integer (if it exists) such that, for all $n \geq n_0$, distinct positive integers $a_1, \ldots, a_r$ exist with

$$f(a_1) + \cdots + f(a_r) = n$$

and

$$\frac{1}{a_1} + \cdots + \frac{1}{a_r} = \alpha.$$

After showing in [1] that $n_0(x, 1)$ is equal to 78 and, more generally, that $n_0(x, \alpha)$ exists for all $\alpha > 0$, Graham furthermore conjectured the existence of $n_0(f(x), \alpha)$ for all positive rationals α, as long as $f(x)$ has positive leading coefficient and $\gcd(f(1), f(2), \ldots) = 1$. Erdős and Graham then repeated this conjecture for $\alpha = 1$ in [2, p. 32], and this special case is now listed as Erdős Problem #283 at [3].

However, besides near-optimal upper bounds for $n_0(x, \alpha)$ that were given in [4] by the author, the only other case that has been solved is $f(x) = x^2$ for $\alpha = 1$; Alekseyev proved in [5] that $n_0(x^2, 1)$ is equal to 8543.

In this paper we will add to this knowledge, and show that if $f(x) = ax^d + b$ is a binomial with coprime coefficients, then for all $\alpha > 0$, a finite computation is potentially sufficient to find the value of $n_0(f(x), \alpha)$. For $a = 1$, $b = 0$ such a result was essentially already obtained in [5, Theorem 6] and [6, Theorem 4], nicely paving the way for the current generalization. As a proof of concept we will then carry out this computation for some specific rationals and (mostly linear) polynomials and, for example, show $n_0(x + b, 1) \leq 172 + 10b$ for all positive integers $b \leq 5000$.

As for our notation, we use the greek letters α and β to denote positive rationals, while the letters A, A_i and B refer to sets of positive integers. For a function $f(x)$ and a set $A = \{a_1, \ldots, a_r\}$ of positive integers, $\sum f(A)$ denotes $f(a_1) + \cdots + f(a_r)$, with in particular $\sum A^{-1} = \frac{1}{a_1} + \cdots + \frac{1}{a_r}$ and $\sum A^d = a_1^d + \cdots + a_r^d$. Finally, for a positive integer m, we write mA for the set $\{ma_1, \ldots, ma_r\}$.

2 Main result, first version

For all positive rationals α and all integer polynomials $f(x) = ax^d + b$ with $a \in \mathbb{N}$ and $\gcd(a, b) = 1$, our main result will essentially reduce the existence and value of $n_0(f(x), \alpha)$ to a finite computation. This theorem will be modeled after Theorem 2 in [6], so familiarity with its statement and proof is definitely helpful, but by no means required[1]. We further refer to [6, Section 4] for a discussion on these types of metatheorems, as well as examples showing their value.

Let S be a set of positive rationals such that for all $\alpha \in S$ we want to prove that for all large enough n, a set A exists for which $\sum f(A) = n$ and $\sum A^{-1} = \alpha$. To prove this, we choose a suitable property Q that a set can have (e.g. not containing certain integers), a fixed cardinality s divisible by a, an integer $m \geq 2$ coprime to a and, for every $\alpha \in S$, rationals $\beta_1, \ldots, \beta_{m^d}$ and sets of positive integers $A_1, \ldots, A_{m^d}$ with the following six properties:

1. For all i, $\beta_i \in S$.
2. For all i, $\sum A_i^d \equiv i \pmod{m^d}$.
3. For all i, $|A_i| = s$.
4. For all i, $\alpha = \sum A_i^{-1} + \frac{\beta_i}{m}$.
5. For all i and for every set B with property Q, we have $A_i \cap mB = \emptyset$.
6. For all i and for every set B with property Q, the union $A_i \cup mB$ also has property Q.

[1] For those indeed familiar with it, the most important difference with the theorem we are about to prove, is that here we need the sets under consideration to all have the same cardinality, due to the non-zero constant term b. However, for any set $A = \{a_1, \ldots, a_r\}$ with cardinality r, we always have $\sum f(A) \equiv br \pmod{a}$. A further annoyance is therefore that we have to deal with all residue classes modulo a separately.

With L, M and T defined as

$$\begin{aligned} L &:= \min_{i,\alpha} \sum A_i^d, \\ M &:= \max_{i,\alpha} \sum A_i^d, \\ T &:= \left\lceil \frac{a(M-L)}{m^d-1} \right\rceil - 1 \end{aligned}$$

respectively, we then have the following theorem.

Theorem 1.[2] *Assume that for all $\alpha \in S$ such β_i and A_i exist, satisfying the above six properties. Further assume that positive integers r and X exist with*

$$br(m^d-1) - bs + a(M-L) \geq 0, \tag{1}$$

$$bs(m^d-1) + a(M-L) \geq 0, \tag{2}$$

and such that for all $\alpha \in S$ and all n with

$$X - T \leq n \leq m^d X + aM + T \quad \text{and} \quad n \equiv br \pmod{a},$$

sets A exist with property Q, all with the same cardinality r, and for which $\sum f(A) = n$ and $\sum A^{-1} = \alpha$. Then for all $\alpha \in S$ and all $n \geq X - T$ with $n \equiv br \pmod{a}$, a set A with property Q exists for which $\sum f(A) = n$ and $\sum A^{-1} = \alpha$.

Proof. For a non-negative integer k, define

$$\begin{aligned} Y_0 &:= X, \\ Z_0 &:= m^d X + aM, \\ U_k &:= b(r+ks)(m^d-1) - bs, \\ Y_{k+1} &:= m^d Y_k + aL - U_k, \\ Z_{k+1} &= m^d Z_k + aM - U_k, \\ I_k &:= [Y_k - T, Z_k + T], \end{aligned}$$

and assume that for all $n \in I_k$ with $n \equiv br \pmod{a}$ sets A exist with property Q, all with the same cardinality $r+ks$, and for which $\sum f(A) = n$ and $\sum A^{-1} = \alpha$. Then by assumption this is true for $k = 0$, and we will prove it true by induction for $k+1$. The proof is then finished if we can show that the union of all intervals I_k contains every integer larger than or equal to $X - T$.

Let an integer $n \in I_{k+1}$ with $n \equiv br \pmod{a}$ be given, and (using the second property) let i be such that

$$n \equiv a \sum A_i^d - U_k \pmod{m^d}.$$

[2]The double checkmark means that the proof was also formalized using the proof assistant Lean. On the author's GitHub page [7] one can find this formalization, which was obtained using the program Aristotle from Harmonic.

This implies that an integer n' exists with which we can write

$$n = m^d n' + a \sum A_i^d - U_k.$$

With this value of n', we claim $n' \in I_k$. Indeed, on the one hand we have

$$\begin{aligned} n' &= \frac{n - a\sum A_i^d + U_k}{m^d} \\ &\geq \frac{Y_{k+1} - T - aM + U_k}{m^d} \\ &= \frac{m^d Y_k + aL - U_k - T - aM + U_k}{m^d} \\ &> Y_k - T - 1. \end{aligned}$$

While on the other hand we have

$$\begin{aligned} n' &= \frac{n - a\sum A_i^d + U_k}{m^d} \\ &\leq \frac{Z_{k+1} + T - aL + U_k}{m^d} \\ &= \frac{m^d Z_k + aM - U_k + T - aL + U_k}{m^d} \\ &< Z_k + T + 1. \end{aligned}$$

Moreover, as a divides s, we have

$$\begin{aligned} n' &= \frac{n - a\sum A_i^d + U_k}{m^d} \\ &\equiv \frac{br + br(m^d - 1)}{m^d} \pmod a \\ &\equiv br \pmod a. \end{aligned}$$

Since

$$n' \in I_k \quad \text{and} \quad n' \equiv br \pmod a,$$

we may apply the induction hypothesis on n', so let B be a set with property Q for which $|B| = r + ks$, $\sum f(B) = n'$ and $\sum B^{-1} = \beta_i$. Then we claim that the union $A := A_i \cup mB$ has property Q, $|A| = r + (k+1)s$, $\sum f(A) = n$ and $\sum A^{-1} = \alpha$.

First, A_i and mB are disjoint by the fifth property above, implying

$$|A| = |A_i| + |mB| = s + (r + ks) = r + (k+1)s,$$

while A has property Q by the sixth property. We further get that the sum $\sum A^{-1}$ of reciprocals is equal to

$$\sum A_i^{-1} + \sum (mB)^{-1} = \sum A_i^{-1} + \frac{\beta_i}{m} = \alpha$$

by the fourth property. And finally, the sum $\sum f(A)$ is equal to

$$\begin{aligned}\sum f(A) &= \sum f(A_i) + \sum f(mB) \\ &= a\sum A_i^d + bs + m^d \sum f(B) - b(r+ks)(m^d-1) \\ &= a\sum A_i^d + m^d n' - U_k \\ &= n.\end{aligned}$$

It remains to show that $\bigcup_{k\geq 0} I_k$ contains every integer larger than or equal to $X - T$. It therefore suffices to show $Y_{k+1} \leq Z_k$ and $Z_{k+1} > Z_k$, in which case the intervals overlap and continue on indefinitely.

To see why $Y_{k+1} \leq Z_k$ holds for $k = 0$, we have

$$\begin{aligned}Y_1 &= m^d X + aL - U_0 \\ &= m^d X + aL - br(m^d - 1) + bs \\ &\leq m^d X + aM \\ &= Z_0,\end{aligned}$$

by inequality (1). By induction we then obtain

$$\begin{aligned}Y_{k+1} &= m^d Y_k + aL - U_k \\ &\leq m^d Z_{k-1} + aL - U_k \\ &= Z_k - aM + U_{k-1} + aL - U_k \\ &= Z_k - a(M-L) - bs(m^d-1) \\ &\leq Z_k,\end{aligned}$$

by inequality (2).

Finally, to prove $Z_{k+1} > Z_k$, we note that $\sum f(A) = n$ is by assumption solvable for some $n < X + a$ with some set A with cardinality r, so that

$$X > n - a = \sum f(A) - a \geq br - a.$$

In particular,

$$Z_0 = m^d X + aM \geq X + a > br.$$

In fact, we more generally claim $Z_k > b(r + ks)$ for all $k \geq 0$. Indeed, by induction,

$$\begin{aligned}Z_{k+1} &= m^d Z_k + aM - U_k \\ &> m^d b(r+ks) - b(r+ks)(m^d-1) + bs \\ &= b(r+(k+1)s).\end{aligned}$$

Using this, we get

$$\begin{aligned}
Z_{k+1} &= m^d Z_k + aM - U_k \\
&> Z_k + (m^d - 1)b(r + ks) + aM - b(r + ks)(m^d - 1) + bs \\
&= Z_k + aM + bs.
\end{aligned}$$

This latter quantity is certainly larger than Z_k is $b \geq 0$. On the other hand, if $b < 0$, then inequality (2) gives

$$bs + aM \geq bs(m^d - 1) + a(M - L) \geq 0$$

as well. $\square$

3 Main result, second version

Theorem 1 only considers a specific residue class $br \pmod{a}$. In order to determine $n_0(f(x), \alpha)$ however, we of course need to take all residue classes modulo a into account. Furthermore, Theorem 1 seems to work just for a single binomial $f(x)$, but on closer inspection one can note that the six properties are all independent of the exact value of b. To do this, we start with one crucial but nearly trivial remark: by defining $g(x) := f(x) + 1$, we see that $\sum f(A) = n$ holds for some set A with cardinality r if, and only if, $\sum g(A) = n + r$.

With the above remark in mind, assume that all conditions of Theorem 1 hold, but with the interval

$$X - T \leq n \leq m^d X + aM + T$$

slightly moved to

$$X + r - T \leq n \leq m^d (X + r) + aM + T.$$

Assume further that inequality (1) holds with b replaced by $b + 1$. Then all conditions still hold with f replaced by g, and X replaced by $X + r$, and we may deduce that for all $\alpha \in S$ and all $n \geq X + r - T$ with $n \equiv (b+1)r \pmod{a}$, a set A exists for which $\sum g(A) = n$ and $\sum A^{-1} = \alpha$.

More generally, assume that all conditions of Theorem 1 hold, but with the interval

$$X - T \leq n \leq m^d X + aM + T$$

widened to

$$(X + l_1 r) - T \leq n \leq m^d (X + l_2 r) + aM + T$$

for some integers l_1, l_2 with $l_1 \leq 0 \leq l_2$. Further assume that inequality (1) still holds after replacing b by $b + l_1$ and $b + l_2$. Then Graham's conjecture holds for all large enough $n \equiv (b + j)r \pmod{a}$ and all polynomials of the form

$f(x) = ax^d + b + j$ with $l_1 \leq j \leq l_2$.

In practice this means that, in order to deduce Graham's conjecture for $f(x) = ax^d + b'$ for all b' coprime to a in some interval around b, it is often sufficient to find suitable m, r, s, X, β_i, A_i for all α and for all residue classes modulo a, but only for the polynomial $ax^d + b$. For completeness' sake, the theorem we then apply is as follows.

Theorem 2. *Assume that for all $\alpha \in S$, rationals β_i and sets A_i exist, satisfying the aforementioned six properties, and let l_1, l_2 be integers with $l_1 \leq 0 \leq l_2$. Further assume that for all residue classes $w \pmod a$ integers r_w and $X_w \geq -l_1 r_w$ exist*[3] *with $br_w \equiv w \pmod a$,*

$$(b + l_1)r(m^d - 1) - (b + l_1)s + a(M - L) \geq 0, \tag{3}$$

$$(b + l_2)r(m^d - 1) - (b + l_2)s + a(M - L) \geq 0, \tag{4}$$

$$(b + l_1)s(m^d - 1) + a(M - L) \geq 0, \tag{5}$$

and such that for all $\alpha \in S$ and all n with

$$X_w + l_1 r_w - T \leq n \leq m^d(X_w + l_2 r_w) + aM + T \qquad \textit{and} \qquad n \equiv w \pmod a,$$

sets A exist with property Q, all with the same cardinality r_w, and for which $\sum f(A) = n$ and $\sum A^{-1} = \alpha$. With $f_j(x) := f(x) + j$ we then have that the integer $n_0(f_j(x), \alpha)$ exists for all $\alpha \in S$ and all integers j with $l_1 \leq j \leq l_2$ and $\gcd(b + j, a) = 1$. Morever, for all such j we then have the upper bound

$$n_0(f_j(x), \alpha) \leq \max_w (X_w + jr_w - T).$$

By comparing inequalities (3), (4) and (5) from Theorem 2 with inequalities (1) and (2) from Theorem 1, we remark that

$$(b + j)s(m^d - 1) + a(M - L)$$

is an increasing function of j, which is why we only have to check inequality (5) for $j = l_1$. Similarly,

$$(b + j)r(m^d - 1) - (b + j)s + a(M - L)$$

is either an increasing or decreasing function of j, depending on the sign of $r(m^d - 1) - s$. Therefore we only have to verify inequalities (3) and (4) at the boundary points. More precisely,

$$(b + j)r(m^d - 1) - (b + j)s + a(M - L) \geq 0$$

is implied by inequality (3) if $r(m^d - 1) - s \geq 0$, and by (4) if $r(m^d - 1) - s \leq 0$. We further remark that, if $b + l_1$ is non-negative and $r(m^d - 1) \geq s$, then inequalities (3), (4) and (5) are all automatically satisfied.

[3] Whenever we apply Theorem 2, we always have $X_1 = X_2 = \cdots = X_a$, and we simply denote the integer equal to all of them by X.

4 Applications

The goal of this section is to apply Theorem 2 in order to conclude that with $\alpha = 1$, Graham's conjecture holds for many linear and quadratic polynomials, at least when a is small. Let us start off with linear polynomials with leading coefficient equal to 1.

Theorem 3. *With $l = 5000$, let $f(x)$ be equal to $x + b$, where b is an integer with $1 \leq b \leq l$. Then $n_0(f(x), \alpha) \leq 172 + 10b$ for all $\alpha \in \{\frac{4}{5}, 1\}$.*

Proof. By Theorem 1 we have to define multiple sets, integers and rationals, so let us do so. We choose $S = \{\frac{4}{5}, 1\}$, we let Q be the property of not intersecting $\{13, 23, 57, 65, 85, 115, 117\}$, and we set $r = 10$, $s = 7$, $m = 2$ and $X = 182$. For $\alpha = \frac{4}{5}$ we set

$$\begin{aligned} \beta_1 &= \frac{4}{5}, \\ A_1 &= \{3, 27, 119, 135, 170, 234, 273\}, \\ \beta_2 &= 1, \\ A_2 &= \{5, 19, 46, 105, 114, 190, 483\}. \end{aligned}$$

For $\alpha = 1$ we set

$$\begin{aligned} \beta_1 &= \frac{4}{5}, \\ A_1 &= \{3, 5, 21, 130, 230, 273, 299\}, \\ \beta_2 &= 1, \\ A_2 &= \{3, 9, 39, 91, 105, 130, 585\}. \end{aligned}$$

One can then verify that all six properties from Theorem 1 hold, where $L = 962$, $L = 961$, $T = 0$. It is by Theorem 2 now sufficient to find sets $A = \{a_1, \ldots, a_r\}$ with

$$(a_1 + 1) + \cdots + (a_r + 1) = n \quad \text{and} \quad \frac{1}{a_1} + \cdots + \frac{1}{a_r} = \alpha$$

with property Q for all $\alpha \in S$ and all n with

$$X \leq n \leq 2(X + lr) + M = 101326.$$

One can find these sets at [7]. $\square$

When $\max(a, c) > 1$, the computational requirements in order to apply Theorem 1 quickly add up. If $c > 1$, then X, m^d and M all generally increase substantially, which greatly increases the length of the interval $X \leq n \leq m^d X + aM$ one has to check. And if $a > 1$, then we need to check all residue classes modulo a separately in order to prove that all sufficiently large n are representable. Every residue class requires a different value of r however, and increasing r magnifies the search space dramatically. That being said, as a proof of concept we tackled

$(a, c) = (5, 1)$ and $(a, c) = (1, 2)$. The reader is cordially invited to write more efficient code (or to perhaps tweak the requirements of Theorem 2) in order to cover (way) more polynomials, but for now we shall leave it at the two cases we mentioned.

Theorem 4. *With $l = 375$, let $f(x)$ be equal to $5x + b$, where $1 \le b < l$ is an integer not divisible by 5. Then $n_0(f(x), \alpha) \le 20000$ for all $\alpha \in \{\frac{4}{5}, 1\}$.*

Theorem 5. *With $l = 800$, let $f(x)$ be equal to $x^2 + b$, where b is an integer with $1 \le b \le l$. Then $n_0(f(x), \alpha) \le 50000$ for all $\alpha \in \{\frac{4}{5}, 1, \frac{6}{5}\}$.*

Proof of Theorem 4. We choose $S = \{\frac{4}{5}, 1\}$, we let Q be the property of not intersecting $\{17, 117, 119, 133, 143\}$, and we set $s = 10$, $m = 2$ and $X = 2270$. As we mentioned before, we need a different value of r for every residue class modulo a, and we choose $r \in \{11, 12, 13, 14, 15\}$.

For $\alpha = \frac{4}{5}$ we set

$$\begin{aligned}
\beta_1 &= 1, \\
A_1 &= \{9, 13, 39, 45, 63, 65, 77, 105, 165, 234\}, \\
\beta_2 &= \frac{4}{5}, \\
A_2 &= \{5, 15, 33, 34, 35, 63, 99, 105, 187, 238\}.
\end{aligned}$$

For $\alpha = 1$ we set

$$\begin{aligned}
\beta_1 &= 1, \\
A_1 &= \{5, 9, 11, 33, 45, 65, 91, 105, 165, 286\}, \\
\beta_2 &= \frac{4}{5}, \\
A_2 &= \{3, 13, 15, 21, 39, 63, 65, 95, 234, 266\}.
\end{aligned}$$

Once again one can then check that all six properties from Theorem 1 hold, with $M - L = 815 - 814 = 1$ and $T = 5$.

At [7] one can now find $A = \{a_1, \dots, a_r\}$ with

$$(5a_1 + 1) + \cdots + (5a_r + 1) = n \quad \text{and} \quad \frac{1}{a_1} + \cdots + \frac{1}{a_r} = \alpha$$

with property Q for all $\alpha \in S$ and all n with

$$X - T \le n \le 2(X + lr) + 5M + T \le 20000. \qquad \square$$

Proof of Theorem 5. We choose $S = \{\frac{4}{5}, 1, \frac{6}{5}\}$ this time around, we let Q be the property of not intersecting $\{13, 17, 19, 39\}$, $r = 11$, $s = 10$, $m = 2$ and $X = 29000$.

For $\alpha = \frac{4}{5}$ we set $\beta_1 = \beta_3 = 1$ and $\beta_2 = \beta_4 = \frac{4}{5}$ with the following sets:

$$\begin{aligned} A_1 &= \{11, 21, 26, 33, 35, 55, 63, 77, 99, 143\} \\ A_2 &= \{7, 9, 23, 45, 55, 63, 69, 77, 99, 115\} \\ A_3 &= \{9, 26, 34, 35, 45, 51, 63, 65, 78, 153\} \\ A_4 &= \{5, 15, 26, 55, 63, 65, 77, 78, 99, 105\}. \end{aligned}$$

For $\alpha = 1$ we set $\beta_1 = \beta_3 = 1$ and $\beta_2 = \beta_4 = \frac{6}{5}$ with the following sets:

$$\begin{aligned} A_1 &= \{5, 9, 15, 21, 55, 77, 78, 91, 99, 105\} \\ A_2 &= \{7, 9, 23, 45, 55, 63, 69, 77, 99, 115\} \\ A_3 &= \{5, 7, 26, 35, 38, 57, 65, 78, 95, 133\} \\ A_4 &= \{5, 15, 26, 55, 63, 65, 77, 78, 99, 105\}. \end{aligned}$$

Finally, for $\alpha = \frac{6}{5}$ we set $\beta_1 = \beta_3 = 1$ and $\beta_2 = \beta_4 = \frac{4}{5}$ with the following sets:

$$\begin{aligned} A_1 &= \{3, 7, 15, 21, 34, 35, 51, 63, 105, 153\} \\ A_2 &= \{3, 5, 7, 21, 55, 65, 77, 91, 99, 117\} \\ A_3 &= \{3, 5, 26, 34, 35, 51, 63, 65, 78, 153\} \\ A_4 &= \{3, 5, 9, 21, 26, 45, 65, 77, 78, 165\}. \end{aligned}$$

We now have $M - L = 46725 - 43379 = 3346$ and $T = 1115$.

At [7] one can find $A = \{a_1, \dots, a_r\}$ with

$$(a_1^2 + 1) + \cdots + (a_r^2 + 1) = n \qquad \text{and} \qquad \frac{1}{a_1} + \cdots + \frac{1}{a_r} = \alpha$$

with property Q for all $\alpha \in S$ and all n with

$$X - T \leq n \leq m^d(X + lr) + M + T \leq 200000. \qquad \square$$

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